ReDance
Sensor-driven neurorehabilitation

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Patient Case

Demographics

- 74 years old
- Mother of 2 Children
- Married

Background

- Dedicated Housewife
- Very involved family

Likes

- Music, Rhythm, dancing
- Cleanliness and Tidiness
- Gymnastics
- Hiking

Dislikes

- Singing
- Crumpled Shirts
- Dirt / Stains
- Exhausting exercise

Clinical movements anchor the routine

Item	Score	Training Movement
9. Pick up object from ground from a standing position	3	Trunk Flexion
10. Turning to look behind over left and right shoulders while standing	2	Trunk Rotation
12. Alternate placing of feet on stair, while standing unsupported	2.5	Hip Flexion

Why gamification?

Repetition becomes engaging and motivating

Meta-Analysis > Biol Res Nurs. 2024 Jan;26(1):5-20. doi: 10.1177/10998004231180316. Epub 2023 May 29.

Wii Fit-Based Biofeedback Rehabilitation Among Post-Stroke Patients: A Systematic Review and Meta-Analysis of Randomized Controlled Trial

Seyedeh Maryam Ghazavi Dozin [1], Nasser Mohammad Rahimi [2], Reza Aminzadeh [1]

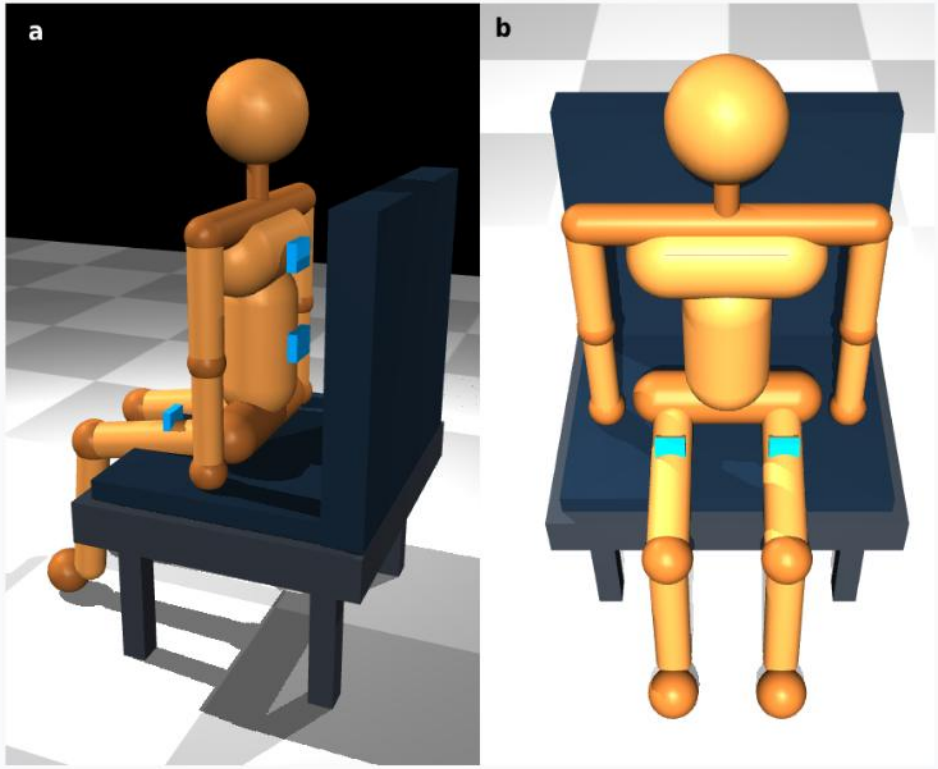
Affiliations + expand

PMID: 37247514 DOI: 10.1177/10998004231180316

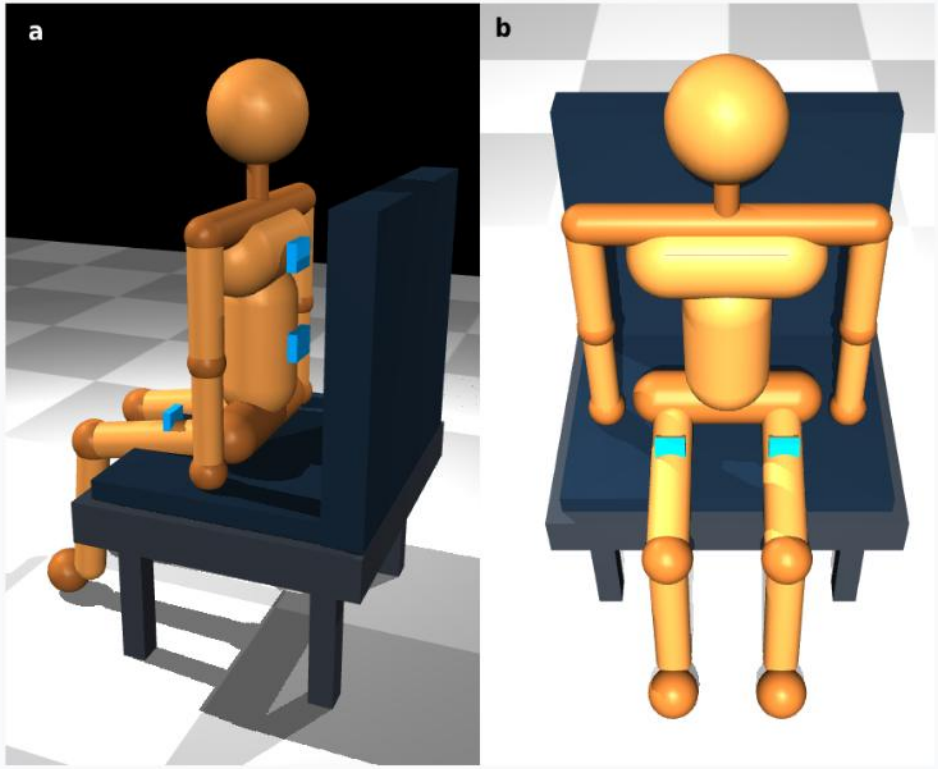
Abstract

Background: Stroke is one of the most widespread reasons for acquired adult disability. Recent

System Overview



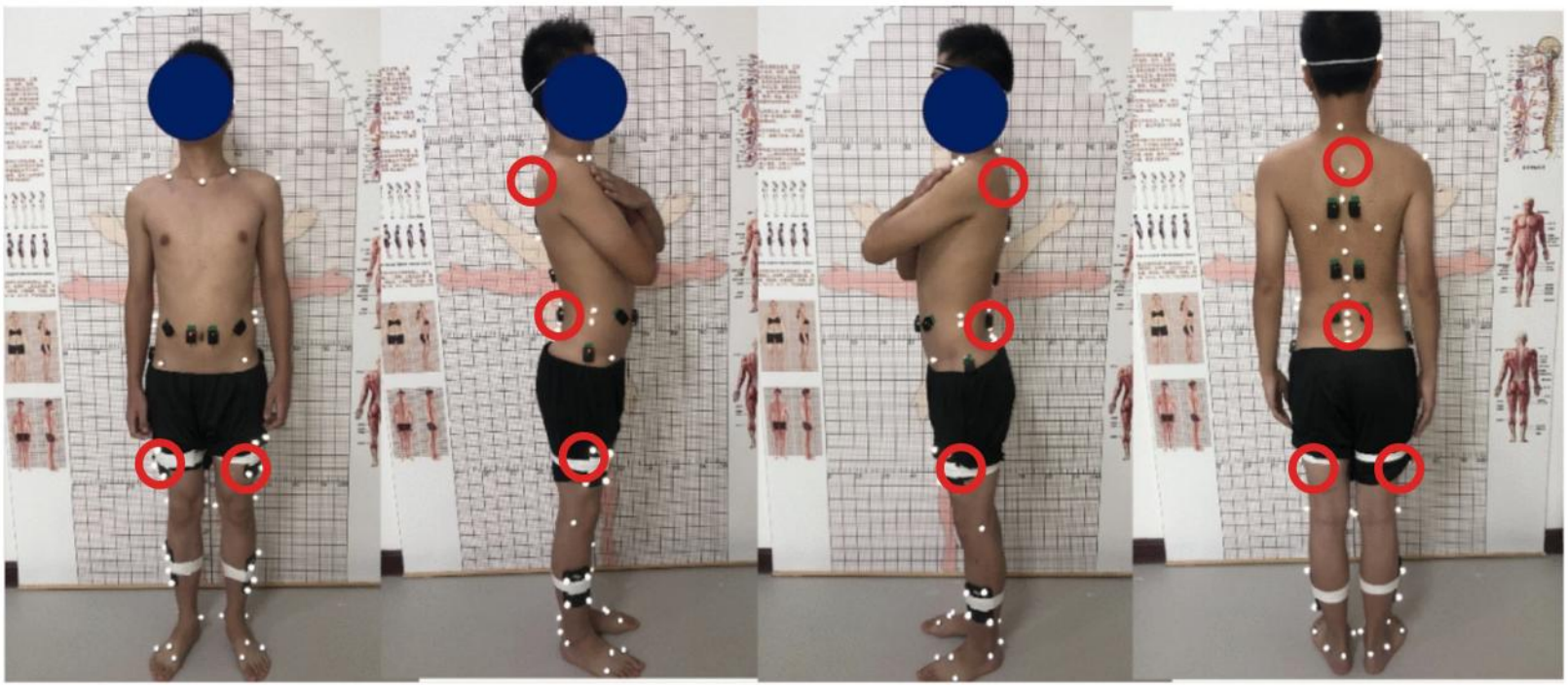
Safety Harness



Five IMUs capture the movement chain

Figure 2

From: Biomechanical analysis of trunk and lower limbs during stair activity in patients with scoliosis



IMU data becomes movement feedback

1. Acquire & calibrate

- Five IMUs stream motion
- Neutral posture sets zero
- Signals are synchronized

2. Derive movement

- Compare sensor angles
- Trunk flexion and rotation
- Left and right hip flexion

3. Classify & log

- Drive live feedback
- Save sessions for replay

One interface, two rehabilitation modes

1. Shared interface

- One movement model
- One avatar and routine
- One session record

2. Live patient mode

- Sensors control the game
- Simple cues and rewards
- Immediate correction

3. Recorded physio mode

- Replay saved sessions
- Inspect angles and compensation

Patient mode: live, simple, motivating

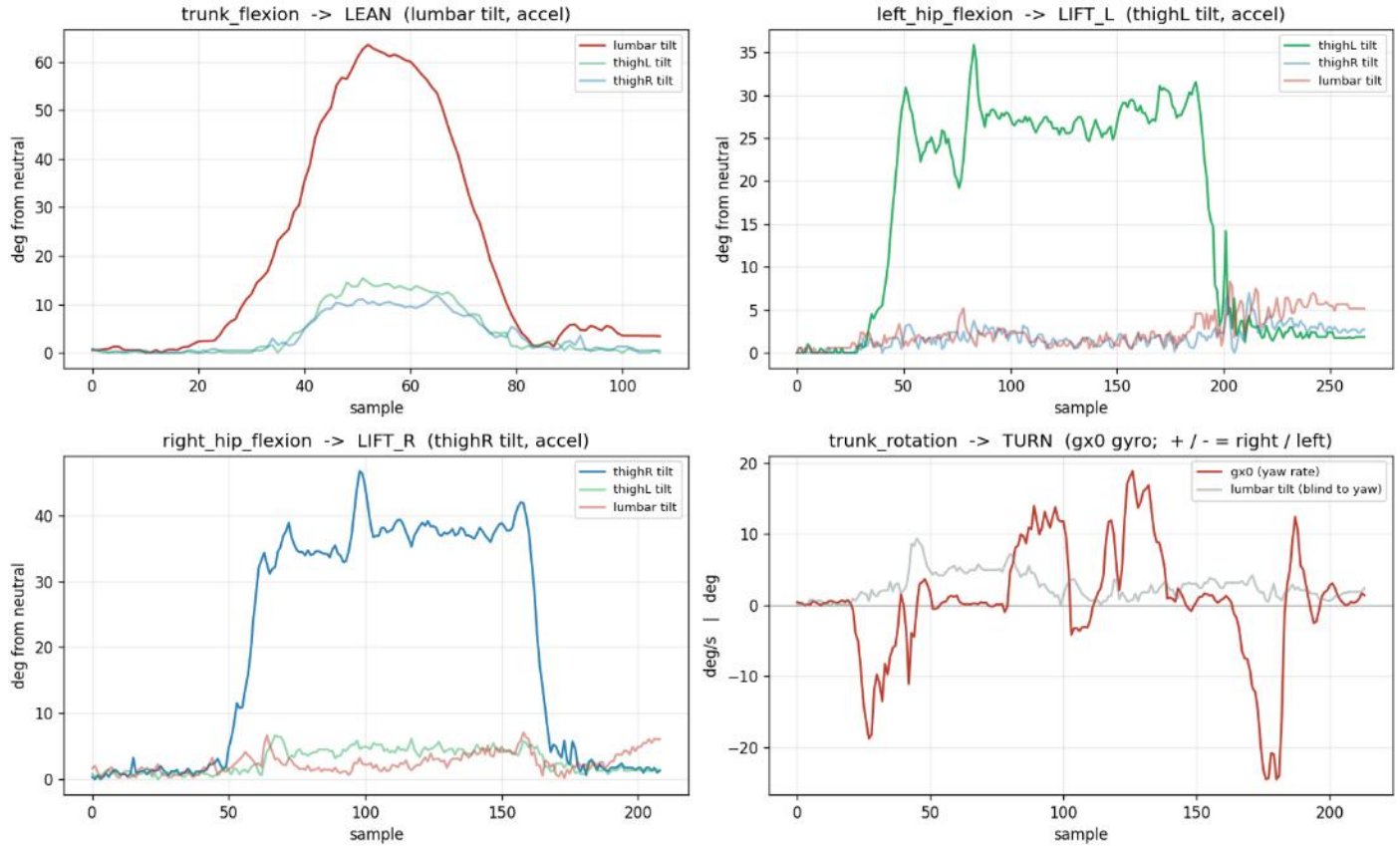
Live patient mode

IMUs control the routine in real time.

Patient-facing feedback

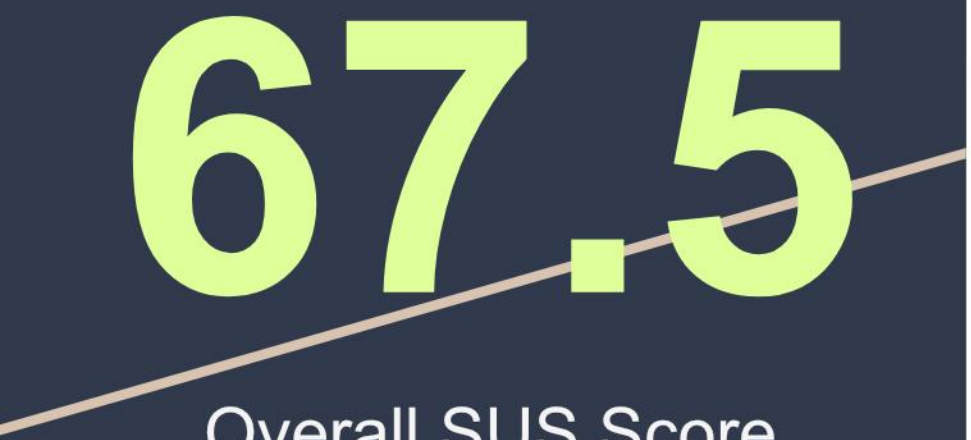
- Current exercise, instruction video, score and progress.
- Movement bars show when thresholds are reached.
- Rewards reinforce correct repetitions.





Physio mode: recorded replay • detailed IMU angle traces

Usability Assessment (SUS)



Solid Foundational Prototype

The score sits just below the industry average of **68.0**, placing the system in the "Marginal" to "Average" category.

- Represents a stable starting point.
- Identifies clear, actionable improvement areas.
- Requires refinement to exceed standard usability benchmarks.

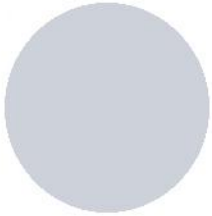
Workload & Friction Analysis

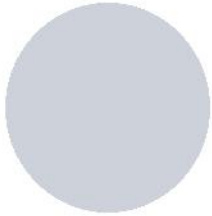
The "Frustration Gap"

While the **zero temporal demand** indicates a comfortable pace, the **maximum levels of effort and frustration** suggest significant mechanical friction.

Next improvements

Simplify the interaction flow, reduce mechanical friction, and surface clear clinician metrics. These changes turn the prototype into a more reliable rehabilitation tool.

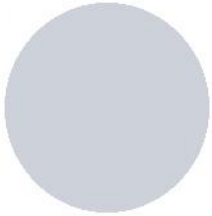


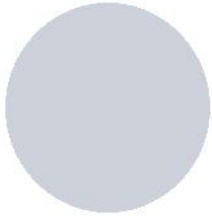


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The team behind ReDance

Hardware • sensing • movement logic •
patient and physio interfaces





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Live demo

From calibration to movement feedback and session replay

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Questions?

Additional movement metrics and plots follow in the appendix

Appendix
Measurement details

Trunk Flexion

1. Difficulty Levels
- Sitting on chair
- Sitting on high chair
- Standing without support

2. Range of Movements
- Reach knees
- Reach ankles
- Reach floor

3. Movement variations
- Unilateral
- Bilateral

Trunk Flexion: How to Measure

1. Sensors processing

- Individual sensor angles
- Sensors angle differences

2. Individual sensor angles

Tilt Lateral

$$\theta = \cos^{-1}\left(\frac{a_z}{\sqrt{a_y^2 + a_z^2}}\right) - \frac{\pi}{2}, \qquad \varphi = \frac{\pi}{2} - \cos^{-1}\left(\frac{a_x}{\sqrt{a_x^2 + a_y^2}}\right)$$

3. Dynamic angle

- Track angle change through time
- Detect cheating and compensation
- Give real-time corrective feedback

Trunk Flexion: Possible Compensation and Cheating

1. Lateral leaning

- Lateral trunk flexion/rotation
- Compare sensors lateral angle
- Train weaker side with unilateral movement

2. Wall leaning

- Not standing unsupported
- Pelvis IMU not static
- Safety sofa

3. Sensor manipulation

- More than one sensors
- Unreachable sensor placement
- Dynamic angle tracking

Trunk Rotation

Primary Drivers

Trunk rotation is a composite movement primarily driven by:

- Thoracic spine
- Lumbar spine

Seated Active Range of Motion (ROM)

40° to 45°

Per direction (relative to neutral sagittal plane)

Total Transverse Arc:

80° to 90° of total rotational capacity.

Asymmetry Threshold:

A difference > 10% to 15% between left and right rotation.

Clinical Significance

Indicates clinically significant mobility deficits or unilateral core muscle weakness when threshold is exceeded.

How to Measure Trunk Rotation

Test Configuration

Positioning: The participant sits upright on a backless chair or bench, feet flat on the floor, with knees and hips flexed at 90°.

Math: If using IMUs, the rotation angle θ_trunk is computed by establishing a vector connecting the two acromion processes (v_shoulder) relative to a fixed pelvic vector (v_pelvis) across the anterior superior iliac spines (ASIS). The angle in the transverse plane is given by:

Setup

IMUs:

1. Two IMUs at the clavicle (left and right)
2. One IMU at the pelvis

$$\theta_{\text{trunk}} = \arccos\left(\frac{v_{\text{shoulder}} \cdot v_{\text{pelvis}}}{\|v_{\text{shoulder}}\|\|v_{\text{pelvis}}\|}\right)$$

Trunk Rotation:

Possible Compensation and Cheating Mechanisms

1. Pelvic and Hip Shifting

Mechanism: Subject slides knee forward or lifts hip, rotating pelvis in transverse plane.
Detection: Detection over Thigh IMUs

2. Shoulder Cheating

Mechanism: Independent shoulder movement (protraction/retraction) when arms/shoulders aren't locked.
Detection: Thoracic spine is static; track T1 sensor relative to acromion markers.

3. Lateral Spine Flexion

Mechanism: Combines rotation with lateral tilt (side-bend) via lumbar coupling.
Detection: Look for sudden deviation in coronal plane (Y-axis roll/velocity).

Hip Flexion
Possible Compensation and Cheating Mechanisms

Beginner Cheating Proof
Ask patient to lift knees as high as possible to measure hip flexion and torso engagement.

Difficulty Levels (Example: 45° Max Engagement)

- **Easy (30 - 43°):** Practice the movement rather than achieving max engagement.
- **Medium (44 - 59°):** Working on gaining more hip mobility and engagement.
- **Hard (60 - 65°):** Solely focused on engaging torso and hips after proper practice.

Compensation & Feedback
Measure thorax/pelvic concavity. Reduce via negative feedback if limits are exceeded.

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